

# Economies of Scale in Artificial Intelligence Innovation

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## Abstract

Artificial intelligence (AI) technologies are increasingly considered general-purpose technologies with the potential to transform production across industries. A defining characteristic of AI systems is their cost structure: development requires substantial fixed investments in data infrastructure, computing resources, and specialized labor, while the marginal cost of deploying these systems is low. This paper empirically examines whether the cost structure of AI development generates scale advantages across firms.

Using firm-level financial data combined with AI-related patent activity, the analysis investigates how AI innovation output varies with firm size. AI innovation is proxied using AI-related patents derived from USPTO data and linked to firm financials from Compustat. A series of regression models, including baseline OLS, controlled specifications with R&D and capital intensities, industry-year fixed effects, nonlinear scaling, and firm fixed effects are estimated to quantify the relationship between firm size and AI innovation.

The results demonstrate that AI patenting increases with firm size. Across baseline and controlled specifications, the elasticity of AI patenting with respect to employment ranges from approximately 0.15 to 0.23. Nonlinear models suggest modest convex scaling, though the overall relationship remains sublinear. Poisson count models yield consistent findings, indicating that larger firms produce more AI patents on average.

# 1 Introduction

Artificial intelligence (AI) technologies have become central to modern production systems. Firms increasingly invest in machine learning models, large-scale data infrastructure, and automated decision systems to improve productivity, enhance products, and expand operational capabilities. Understanding the economic structure of AI investment is critical, given AI’s potential to reshape firm capabilities and industry dynamics.

A distinctive feature of AI technologies is their cost structure. Developing an AI system requires substantial upfront investment, including data collection, model training, specialized computing infrastructure, and highly skilled labor. Once developed, however, deploying the AI system across additional tasks or users entails relatively low marginal costs. This pattern is analogous to other digital technologies and suggests that AI may generate significant economies of scale.

Firms with greater size can potentially spread the fixed costs of AI development across more employees or operations, gaining cost advantages over smaller competitors. Conversely, smaller firms may face higher average costs, which could constrain their capacity to develop AI innovations.

This paper investigates whether AI innovation activity scales with firm size and whether this relationship exhibits nonlinear characteristics. Using firm-level data from Compustat and AI-related patent data from the USPTO, the analysis examines how AI patent output varies with firm employment and whether larger firms exhibit stronger scaling effects.

Empirical results indicate three main findings. First, AI patenting systematically increases with firm size. Second, the elasticity of AI patenting with respect to employment ranges from 0.15 to 0.23 across baseline specifications. Third, nonlinear models suggest modest convex scaling patterns, implying somewhat stronger innovation responses among larger firms.

## 2 Related Literature

This study contributes to research on technology adoption, firm-level innovation, and AI economics.

A substantial literature examines how fixed costs influence technology adoption. Models of heterogeneous firms indicate that larger or more productive firms are more likely to adopt technologies with high fixed costs (Acemoglu and Atkeson, 2010). Adoption occurs when firms can recover fixed costs through sufficiently large production scales.

Empirical research on IT investment demonstrates similar scale effects. Studies on enterprise computing and digital infrastructure find that larger firms benefit from spreading fixed costs across greater output (Brynjolfsson and Hitt, 2000; Bartel et al., 2007). These findings imply that digital technologies frequently exhibit increasing returns to scale at early adoption stages.

Emerging research on AI examines its impact on productivity, labor demand, and automation (Brynjolfsson and McAfee, 2021; Agrawal et al., 2019). However, empirical work directly assessing how AI development scales with firm size remains limited. This paper provides new evidence by linking firm characteristics to AI patenting outcomes.

### 3 Conceptual Framework

AI development can be described with a simple cost model:

$$C = F + cQ \tag{1}$$

where  $F$  represents fixed development costs,  $c$  represents marginal deployment costs, and  $Q$  is the scale of deployment.

Average cost is therefore:

$$AC = \frac{F}{Q} + c \tag{2}$$

As scale increases, the fixed-cost component declines, giving larger organizations lower average costs per unit of AI innovation. This framework predicts that larger firms will produce more AI output per employee and exhibit scale advantages.

## 4 Data

### 4.1 Data Sources

The dataset combines firm-level financial information from Compustat with AI-related patent counts from the USPTO. The panel covers multiple years and includes firm identifiers (GVKEY) and industry classifications (NAICS codes).

### 4.2 Variable Construction

Firm size is measured using the logarithm of employment (`log_emp`). AI innovation output is proxied by AI-related patent counts (`ai_patent_count`), log-transformed as  $\log(1 + AI)$ .

Control variables include:

- R&D intensity: R&D expenditures divided by revenue (`rd_intensity`)
- Capital intensity: property, plant, and equipment divided by total assets (`capital_intensity`)
- Capital expenditure intensity: capital expenditures divided by total assets (`capx_intensity`)

Industry (2-digit NAICS) and year fixed effects are included in later models. All intensity variables are winsorized at the 1st and 99th percentiles to limit outlier influence.

### 4.3 Summary Statistics

Table 1: Summary Statistics

| Variable          | Mean  | Std. Dev. | Min   | Max    |
|-------------------|-------|-----------|-------|--------|
| Employment        | 49.89 | 94.20     | 0.006 | 671.69 |
| AI Patents        | 6.01  | 14.88     | 0     | 135    |
| R&D Intensity     | 0.08  | 0.08      | 0     | 0.41   |
| Capital Intensity | 0.12  | 0.15      | 0.01  | 0.97   |
| Capex Intensity   | 0.03  | 0.04      | 0     | 0.25   |

## 5 Empirical Strategy

This study examines the relationship between firm size and AI innovation using a sequence of increasingly restrictive regression specifications. The dependent variable in all models is  $\log(1 + AI_{it})$ , where  $AI_{it}$  denotes the number of AI-related patents produced by firm  $i$  in year  $t$ . Firm size is measured by total employment,  $Emp_{it}$ .

The baseline specification estimates the relationship between AI patenting and firm size:

$$\log(1 + AI_{it}) = \alpha + \beta_1 \log(Emp_{it}) + \epsilon_{it} \quad (3)$$

The next specification incorporates firm-level controls:

$$\log(1 + AI_{it}) = \alpha + \beta_1 \log(Emp_{it}) + X_{it}\gamma + \epsilon_{it} \quad (4)$$

where  $X_{it}$  includes `rd_intensity`, `capital_intensity`, and `capx_intensity`.

To account for industry and time effects, the following specification is estimated:

$$\log(1 + AI_{it}) = \alpha + \beta_1 \log(Emp_{it}) + X_{it}\gamma + \delta_j + \lambda_t + \epsilon_{it} \quad (5)$$

The analysis then allows for nonlinear scaling by introducing a quadratic term in log employment:

$$\log(1 + AI_{it}) = \alpha + \beta_1 \log(Emp_{it}) + \beta_2(\log(Emp_{it}) - \overline{\log(Emp)})^2 + X_{it}\gamma + \delta_j + \lambda_t + \epsilon_{it} \quad (6)$$

Centering the log employment variable allows  $\beta_2$  to capture convexity in the scaling relationship while reducing collinearity between the linear and quadratic terms.

Finally, a firm fixed-effects specification is estimated:

$$\log(1 + AI_{it}) = \alpha + \beta_1 \log(Emp_{it}) + \beta_2(\log(Emp_{it}) - \overline{\log(Emp)})^2 + X_{it}\gamma + \mu_i + \lambda_t + \epsilon_{it} \quad (7)$$

This specification controls for all time-invariant firm characteristics and identifies the relationship using within-firm variation over time.

## 6 Results

Table 2 presents a summary of regression results across the five main specifications. These results provide a clear overview of how AI patenting responds to firm size under different model configurations.

- **Baseline OLS:** The elasticity of AI patenting with respect to employment is 0.15, indicating a positive relationship between firm size and innovation.
- **OLS with Controls:** Including R&D, capital, and capex intensities increases the elasticity to 0.18.
- **Industry-Year FE:** Adding fixed effects further increases the elasticity to 0.23.
- **Nonlinear Model:** Coefficients on the squared centered log employment variable are positive, suggesting modest convex scaling. Evaluated at the mean firm size, the marginal elasticity is approximately 0.27.
- **Firm Fixed Effects Nonlinear Model:** Firm fixed effects reduce the statistical significance of the employment coefficient, suggesting that much of the observed relationship operates across firms rather than through within-firm changes over time.

Table 2 provides a concise representation of these findings, summarizing the estimated elasticities and notes for each specification.

Table 2: Regression Summary of AI Patents on Firm Size

| Specification     | Elasticity (log emp) | Notes                            |
|-------------------|----------------------|----------------------------------|
| Baseline          | 0.15                 | OLS, clustered SE                |
| Controls          | 0.18                 | OLS + R&D, capital, capex        |
| Industry-Year FE  | 0.23                 | OLS + fixed effects              |
| Nonlinear         | 0.27                 | Marginal elasticity at mean size |
| Firm FE Nonlinear | 0.27                 | Firm and year fixed effects      |

To provide a more detailed view of the estimated coefficients across all model specifications, Table 3 reports the full regression estimates, including control variables, nonlinear terms, and fixed effects. This table complements the summary by illustrating the underlying parameter estimates that drive the observed elasticities.

Table 3: Firm Size and AI Patent Production

| Variable                       | Baseline            | Controls            | Industry-Year FE    | Nonlinear           | Firm FE Nonlinear  |
|--------------------------------|---------------------|---------------------|---------------------|---------------------|--------------------|
| log(emp)                       | 0.147***<br>(0.029) | 0.180***<br>(0.030) | 0.235***<br>(0.042) |                     |                    |
| log(emp) centered              |                     |                     |                     | 0.270***<br>(0.044) | 0.125<br>(0.187)   |
| log(emp) <sup>2</sup> centered |                     |                     |                     | 0.032***<br>(0.010) | 0.094*<br>(0.056)  |
| rd_intensity                   |                     | 0.221***<br>(0.082) | 0.218***<br>(0.081) | 0.074<br>(0.056)    | -0.076<br>(0.191)  |
| capital_intensity              |                     | -0.920*<br>(0.539)  | -0.784<br>(0.510)   | -1.099**<br>(0.519) | 1.303<br>(1.550)   |
| capx_intensity                 |                     | 3.309<br>(3.018)    | 3.426<br>(2.661)    | 3.980<br>(2.645)    | -4.585*<br>(2.364) |
| Observations                   | 817                 | 817                 | 817                 | 817                 | 817                |
| R <sup>2</sup>                 | 0.108               | 0.126               | 0.252               | 0.275               | 0.772              |

Standard errors clustered at the firm level in parentheses. \*  $p < 0.10$ , \*\*  $p < 0.05$ , \*\*\*  $p < 0.01$ .

To visualize the relationship between firm size and AI patenting, Figure 1 presents a scatter plot of log AI patents against log employment, along with the fitted regression line. This figure illustrates the positive association documented in the regression analyses and provides an intuitive depiction of the scaling relationship.



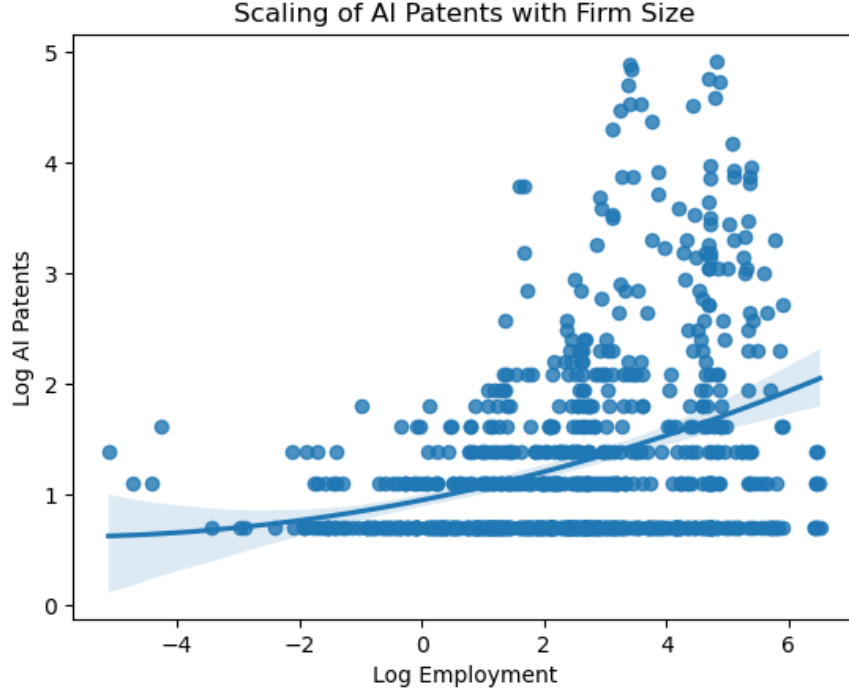


Figure 1: Scaling of AI Patents with Firm Size. Scatter plot with regression line showing the positive relationship between firm size and AI patenting.

## 7 Robustness

Poisson regression models, which account for the count nature of patent data, produce qualitatively similar results to the OLS specifications. Because Poisson models are estimated in levels rather than log-transformed counts, coefficients are not directly comparable to log-log OLS elasticities but confirm the positive relationship between firm size and AI patenting.

Table 4: Poisson Regression of AI Patents on Firm Size (Selected Variables)

| Variable          | Coefficient | Std. Error | Significance |
|-------------------|-------------|------------|--------------|
| Log Employment    | 0.621       | 0.085      | ***          |
| R&D Intensity     | 0.684       | 0.172      | ***          |
| Capital Intensity | -0.741      | 1.413      |              |
| Capex Intensity   | 6.176       | 4.036      |              |

## 8 Discussion and Limitations

Several limitations should be noted when interpreting the results. First, patent counts capture only a subset of firms’ innovation activities. Some AI innovations may be implemented internally without being patented, meaning the measure of innovation may underestimate true technological development. However, patent data remain one of the most widely used proxies for technological output in empirical innovation research.

Second, the analysis relies on observational firm-level data and therefore cannot fully establish causal relationships. While the regression specifications include industry and time fixed effects as well as firm-level controls, unobserved factors may still influence both firm size and innovation activity. The results should therefore be interpreted primarily as evidence of a strong empirical association rather than a definitive causal effect.

Third, measurement of AI innovation depends on classification of patent technologies. Although the patent matching procedure follows established methods in the literature, classification errors or evolving definitions of AI technologies may introduce noise into the measurement of innovation outcomes.

Taken together, these limitations underscore the need for caution in interpreting the results, yet they do not diminish the substantive insights provided by the empirical analysis. The consistent positive relationship between firm size and AI patent production across multiple model specifications suggests that larger firms play a significant role in the development of artificial intelligence technologies.

The results consistently indicate that AI innovation scales with firm size. Nonlinear specifications suggest modest convex scaling, though the overall relationship remains sublinear. Firm fixed effects models indicate that much of the variation in AI patenting occurs across firms rather than within firms over time.

These findings align with the cost structure of AI development, characterized by high fixed costs and low marginal costs. Larger firms can more efficiently absorb development costs, supporting higher innovation output relative to smaller firms.

## 9 Conclusion

This study examines the scaling of AI innovation with firm size using linked firm-level financial data and AI patent counts.

Results demonstrate that AI patenting increases with firm size. While the relationship remains less than proportional, larger firms consistently produce more AI patents on average. These findings support the hypothesis that AI development’s cost structure generates scale advantages that may favor larger firms.

Future research could examine the dynamics of these scale advantages over time, as AI technologies mature and adoption spreads to smaller firms.

## References

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### Recent Revisions (following comments from Dr. David Arnold)

Recent econometric literature has highlighted potential limitations of using  $\log(1 + x)$  transformations when outcome variables contain a substantial number of zero observations. In response, this paper considers the implications of these findings and references the following work:

- Roth, J. (2024). *Logarithms with Zeros*. Available at: Logarithms with Zeros.